

Optimization Loss Function Formulation

Based on the weights and constraints outlined, here is the objective loss function formulated for the optimization setup.

To ensure the 40/20/20/20 percentage weightings function properly without being skewed by differing units (e.g., kilograms vs. liters), it is highly recommended to normalize the variables against baseline or target reference values.

The Loss Equation

$$\mathcal{L} = 0.4 \left(\frac{M}{M_{ref}} \right) + 0.2 \text{ReLU} \left(\frac{LD_{target} - LD}{LD_{target}} \right) + 0.2 \text{ReLU} \left(\frac{V_{f,target} - V_f}{V_{f,target}} \right) + 0.2 \text{ReLU} \left(\frac{V_{p,target} - V_p}{V_{p,target}} \right) + \mathcal{P}_{stress}$$

Equation Breakdown

- **Mass (M):** You want this as low as possible. By dividing by a reference mass (M_{ref}), it scales nicely with the 0.4 multiplier. It scales down as mass decreases.
- **Target Metrics (LD, V_f, V_p):** Using the ReLU function ($\text{ReLU}(x) = \max(0, x)$) perfectly captures your requirement.
 - If the actual value *exceeds* the target, the term inside becomes negative, and the ReLU outputs **0** (no penalty).
 - If the actual value is *below* the target, the term is positive, and the ReLU outputs a proportional penalty weighted by **0.2**.
- **Stress Hard Constraint ($\mathcal{P}_{stress}$):** Since exceeding maximum allowable stress renders the design entirely invalid, it functions as a step-function penalty:

$$\mathcal{P}_{stress} = \begin{cases} 0 & \text{if } \sigma \leq \sigma_{max} \\ \infty & \text{if } \sigma > \sigma_{max} \end{cases}$$

(Note: If your specific optimization algorithm cannot handle infinity, you can replace ∞ with a massive penalty barrier such as $\lambda \text{ReLU}(\sigma - \sigma_{max})$ where λ is an arbitrarily high number like 10^6 .)